Supplementary Information: Genome-Scale Context, Stoichiometric Refinement, and Neuro-Symbolic Reasoning for Proteome Function Annotation

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S1 Hyperparameter specification

All 18 configurable pipeline hyperparameters across Stage 1 (13 `theta.json` keys exposed by `gspa-cli/src/main/groovy/gspa/cl` and consumed by `IterativeRefiner.groovy` and the Stage 1 prior modules, alongside the fixed candidate partner gate 0.08 in `CoherencePrior.groovy`, the fixed co-operonic complex partner bonus `complexOperonPartnerWeight` = 0.95 in `GenomicContextPrior.groovy`, the Stage 2A ancestor relaxation factor `relaxFactor` = 1.0 in `ConsistencyEnforcer.gr` and the Stage 2A obligate-complex singleton prune threshold 0.45 in `IntegrateCommand.groovy`) and Stage 2B (5 parameters in `scripts/stoichiometric_locus_refiner.py`, alongside the fixed elemental-imbalance penalty $w_{\text{imb}} = 100.0$) were calibrated exclusively on the *Escherichia coli* K-12 MG1655 development proteome and held frozen across all out-of-sample prokaryotic, archaeal, viral, and eukaryotic benchmarks. Supplementary Table S1 reports the complete inventory of these 18 configurable hyperparameters, their exact configuration keys, source files, roles in inference, and default values (where penalty parameters `alpha_complex_penalty` = 0.75, `alpha_cons` = 3.00, and `alpha_anat_penalty` = 3.50 are stored as positive magnitudes and applied with negative sign in log-odds space).

S2 Cross-ontology taxonomic and anatomical constraint bridge

Supplementary Table S2 summarizes the five resource tables programmatically extracted from the official OBO Foundry releases of the Gene Ontology (GO), Uberon multi-species anatomy ontology, Plant Ontology (PO), Cell Ontology (CL), and NCBI Taxonomy via `scripts/extract_anatomical_and_taxon_constraints.py` and loaded at runtime by `TaxonConstraints.groovy`, `AnatomicalConstraints.groovy`, and `SatConsistencyChecker.groovy` (`gspa-core/src/main/resources/taxon-constraints/` and `gspa-core/src/main/resources/anatomical-constraints/`).

Table S1. Complete inventory of the 18 configurable GSPA v2.1.0 hyperparameters across `IntegrateCommand.groovy` (`theta.json`) and `scripts/stoichiometric_locus_refiner.py`.

Stage / Module	Symbol	Parameter Key	Source File	Role in Inference	Default Value
Stage 1 (B_1)	α_{ess}	`alpha_ess`	`EssentialityPrior.groovy`	Unfulfilled essential GO function log-odds boost	1.50
Stage 1 (B_2)	α_{coh}	`alpha_coh`	`CoherencePrior.groovy`	Active pathway coherence log-odds scale	1.00
Stage 1 (B_2)	w_{part}	`alpha_complex_boost`	`CoherencePrior.groovy`	Obligate heteromeric complex partner boost ($0.65\,w_{\text{part}} = 0.5525$ for $N_{\text{strong}} \geq 2$)	0.85
Stage 1 (B_2)	w_{sing}	`alpha_complex_penalty`	`CoherencePrior.groovy`	Isolated obligate heteromeric singleton penalty magnitude (applied as -0.75)	0.75
Stage 1 (B_3)	α_{cons}	`alpha_cons`	`ConsistencyPrior.groovy`	SAT4J taxonomic conflict penalty magnitude (applied as -3.00)	3.00
Stage 1 (B_4)	α_{anat}	`alpha_anat_penalty`	`AnatomicalPrior.groovy`	Taxon-forbidden Uberon/PO/CL structure penalty magnitude (applied as -3.50)	3.50
Stage 1 (B_4)	α_{boost}	`alpha_anat_boost`	`AnatomicalPrior.groovy`	Active-structure developmental rescue log-odds boost	0.65
Stage 1 (B_5)	α_{gap}	`alpha_gap`	`GapFillingPrior.groovy`	Metabolic pathway hole-filling log-odds boost	1.50
Stage 1 (B_6)	α_{ctx}	`alpha_ctx`	`GenomicContextPrior.groovy`	Co-operonic pathway enrichment log-odds scale	1.00
Stage 1 (B_6)	$\alpha_{\text{gap_ctx}}$	`alpha_gap_ctx`	`GenomicContextPrior.groovy`	Co-operonic metabolic gap multiplier on $\alpha_{\text{ctx}}\, c_O(W)$	1.50
Stage 1 (Iter)	γ_{iter}	`damping`	`IterativeRefiner.groovy`	Iterative log-odds fixed-point damping weight	0.50
Stage 1 (Iter)	$K_{\max}$	`max_iter`	`IterativeRefiner.groovy`	Maximum Stage 1 message-passing iterations	6
Stage 1 (Iter)	ϵ	`epsilon`	`IterativeRefiner.groovy`	Stage 1 probability convergence tolerance	0.005
Stage 2B (SLR)	τ_{core}	`tau_core`	`stoichiometric_locus_refiner.py`	Core reaction probability threshold $S_{\text{rxn}}(j)$	0.50
Stage 2B (SLR)	α_{bio}	`alpha_bio`	`stoichiometric_locus_refiner.py`	Minimum required fraction of $v_{\text{bio}}^{\max}$ in pFBA	0.10
Stage 2B (SLR)	η_{null}	`tau_null`	`stoichiometric_locus_refiner.py`	Unassigned-gene hypothesis $h_{\varnothing}$ prior mass	0.05
Stage 2B (SLR)	γ_{FBA}	`gamma_fba`	`stoichiometric_locus_refiner.py`	Backward locus responsibility log-odds scale	2.20
Stage 2B (SLR)	γ_{damp}	`gamma_damp`	`stoichiometric_locus_refiner.py`	Geometric damping decay across outer passes	0.60

Table S2. Summary of programmatic cross-ontology constraint tables in GSPA v2.1.0 (`gspa-core/src/main/resources/taxon-constraints/` and `gspa-core/src/main/resources/anatomical-constraints/`).

Resource File	Ontologies Bridged	Data Rows	Distinct Classes	Constraint Semantics & Examples
`go-taxon-constraints.tsv`	GO ↔ NCBITaxon	26,911	18,159 GO terms	13,819 `only_in_taxon` + 13,092 `never_in_taxon` rules (e.g., `GO:0009579` thylakoid)
`go-anatomical-constraints.tsv`	GO ↔ Uberon / PO / CL	2,715	2,674 GO; 1,294 Anat.	30 object properties (`results_in_development_of`, `occurs_in`, etc.)
`uberon-po-taxon-constraints.tsv`	Uberon / PO / CL ↔ NCBITaxon	8,331	1,247 Anat. classes	4,767 `only_in_taxon` + 3,564 `never_in_taxon` rules (e.g., `UBERON:0000948` heart → Metazoa)
`uberon-po-hierarchy.tsv`	Uberon / PO / CL DAG	7,722	2,855 Anat. classes	Transitive `is_a` and `part_of` edges across anatomical structures
`ncbi-taxon-hierarchy.tsv`	NCBI Taxonomy DAG	1,818	1,305 Taxon clades	Transitive taxonomic `is_a` lineage closure for benchmark clades

S3 Extended Suite 1 and Suite 2 benchmarks

Supplementary Table S3 reports the complete 16-row post-cutoff Swiss-Prot enzyme benchmark (`Holdout-125`, $n = 125$ proteins across 293 ground-truth protein–EC pairs; source: `results/scenario1_swissprot_ec_holdout125_real.tsv`), separating standalone per-protein baselines, single-predictor refinement (`DeepProZyme + METEOR v8` and `DeepProZyme + GSPA Single-Pred`), and 3-method ensemble pooling (`Noisy-OR` and `GSPA Iter 1-3`). Because these 125 proteins are dispersed across 74 distinct species and evaluated as isolated sequences, single-predictor and ensemble refinement raise Top-3 recall to 43.2% (54/125) and Top-5 recall to 44.0% (55/125) while preserving Top-1 accuracy (32.0%–33.6% vs. 32.8% for standalone CLEAN). Supplementary Table S4 reports the complete 12-row enzyme evaluation across the 95-genome bacterial panel under four donor-database homology-masking tiers (`vanilla`, $\leq 70\%$, $\leq 50\%$, and $\leq 30\%$ sequence identity; source: `results/scenario1_95genomes_ec_homology_real.tsv`, generated by `scripts/eval_scenario1_holdouts_real.py:340-375`), comparing raw METEOR v8 top-1 flips against GSPA's single-predictor stoichiometric/pathway acceptance filter (`GSPA Single-Pred`, `blind_single_accept`: accepts a candidate METEOR flip $b_1 \to m_1$ when m_1 maps to at least one elementally balanced universal reaction and either shares the 3-digit EC sub-subclass with b_1 at $\geq$ KEGG pathway count or shares the 1-digit EC class with strictly higher KEGG pathway count and $\geq$ balanced reaction count) and GSPA's 3-channel consensus acceptance filter (`GSPA Multi-Pred`, `blind_accept_3ch`: accepts a candidate flip when m_1 receives strictly greater support than b_1 across the other two predictors' base and METEOR top-1 calls), alongside exact discordant protein counts (b_{01}, b_{10}) and two-sided exact McNemar p-values.

S4 Extended stoichiometric audit, curated BiGG reference GEMs, and BacDive cross-validation

Supplementary Table S5 reports the 108-genome stoichiometric invariant audit (`results/pillar2_invariant_audit_summary.ts` the evaluation across six manually curated BiGG reference genome-scale metabolic models (`iML1515` *E. coli*, `STM_v1_0` *S. enterica*, `iYL1228` *K. pneumoniae*, `iJN1463` *P. putida*, `iYS854` *S. aureus*, and `iYO844` *B. subtilis*; source: `results/pillar2_curated6_gems_real.tsv`), and the role-stratified `Price-149` breakdown on CLEAN across Stratum A ($n = 91$), Stratum B ($n = 55$), and Stratum C ($n = 2$; source: `results/pillar2_price149_by_stratum.tsv`). Supplementary Table S6 reports the per-fold 5-fold genus-blocked cross-validation metrics in `BacDive` (source: `results/pillar3_bacdive_per_fold_cv.tsv`).

Table S3. Complete post-cutoff Swiss-Prot enzyme evaluation (`Holdout-125`, $n = 125$ proteins, 293 ground-truth EC annotations; source: `results/scenario1_swissprot_ec_holdout125_real.tsv`).

Predictor / Refinement Method	Masking Tier	Top-1 Acc. (Hits)	Top-3 Acc. (Hits)	Top-5 Acc. (Hits)	MRR	Micro $F_{\max}$
CLEAN (2023)	`vanilla`	32.8% (41/125)	40.8% (51/125)	41.6% (52/125)	0.3719	0.4360
DeepProZyme (2023)	`vanilla`	23.2% (29/125)	31.2% (39/125)	32.0% (40/125)	0.2750	0.2715
DeepProZyme + METEOR v8 (Single-Pred)	`vanilla`	25.6% (32/125)	31.2% (39/125)	31.2% (39/125)	0.2872	0.2804
DeepProZyme + GSPA Single-Pred (SLR + Coh.)	`vanilla`	23.2% (29/125)	31.2% (39/125)	32.8% (41/125)	0.2779	0.2703
EnzBERT (2023)	`vanilla`	32.8% (41/125)	37.6% (47/125)	39.2% (49/125)	0.3531	0.3596
Noisy-OR (3-Method Ensemble)	`vanilla`	33.6% (42/125)	40.8% (51/125)	43.2% (54/125)	0.3807	0.4381
GSPA (3-Method, Iter 1)	`vanilla`	32.0% (40/125)	42.4% (53/125)	44.0% (55/125)	0.3775	0.4035
GSPA (3-Method, Iter 2)	`vanilla`	32.0% (40/125)	42.4% (53/125)	44.0% (55/125)	0.3775	0.3891
GSPA (3-Method, Iter 3)	`vanilla`	32.0% (40/125)	43.2% (54/125)	44.0% (55/125)	0.3781	0.3830
EnzBERT	`filt70` ($\leq 70\%$)	30.4% (38/125)	35.2% (44/125)	35.2% (44/125)	0.3305	0.3204
EnzBERT	`filt50` ($\leq 50\%$)	28.0% (35/125)	33.6% (42/125)	33.6% (42/125)	0.3040	0.3163
EnzBERT	`filt30` ($\leq 30\%$)	17.6% (22/125)	24.8% (31/125)	24.8% (31/125)	0.2139	0.2212
CLEAN	`filt30` ($\leq 30\%$)	24.0% (30/125)	30.4% (38/125)	32.8% (41/125)	0.2741	0.2299
DeepProZyme	`filt70` ($\leq 70\%$)	23.2% (29/125)	31.2% (39/125)	32.0% (40/125)	0.2713	0.2715
DeepProZyme	`filt50` ($\leq 50\%$)	23.2% (29/125)	31.2% (39/125)	32.8% (41/125)	0.2720	0.2684
DeepProZyme	`filt30` ($\leq 30\%$)	22.4% (28/125)	31.2% (39/125)	32.8% (41/125)	0.2674	0.2650

Table S4. Single-predictor and 3-channel consensus acceptance filtering of candidate METEOR v8 EC flips across 95 bacterial genomes under four donor-database homology-masking tiers (source: `results/scenario1_95genomes_ec_homology_real.tsv`; generator: `scripts/eval_scenario1_holdouts_real.py:340-375`). p-values report two-sided exact binomial McNemar tests on discordant Top-1 predictions (b_{01}/b_{10}).

Predictor	Tier	Genomes / Proteins	Base Top-1 ($F_{\max}$)	METEOR v8 Top-1 (Δ)	METEOR b_{01}/b_{10} (p)	GSPA Single-Pred Top-1 (Δ)	Single-Pred b_{01}/b_{10} (p)	GSPA Multi-Pred Top-1 (Δ)	Multi-Pred b_{01}/b_{10} ($F_{\max}$)
CLEAN	`vanilla`	73/52,814	0.8214 (0.8796)	0.8234 (+108)	184/76 ($< 10^{-6}$)	0.8228 (+77)	99/22 ($< 10^{-6}$)	0.8243 (+155)	162/7 (0.8816)
CLEAN	$\leq 70\%$	95/69,663	0.8267 (0.8836)	0.8289 (+149)	191/42 ($< 10^{-6}$)	0.8283 (+108)	131/23 ($< 10^{-6}$)	0.8289 (+149)	154/5 (0.8851)
CLEAN	$\leq 50\%$	95/69,663	0.8271 (0.8844)	0.8294 (+162)	208/46 ($< 10^{-6}$)	0.8291 (+139)	156/17 ($< 10^{-6}$)	0.8293 (+155)	158/3 (0.8859)
CLEAN	$\leq 30\%$	94/68,681	0.8288 (0.8833)	0.8306 (+122)	156/34 ($< 10^{-6}$)	0.8302 (+93)	107/14 ($< 10^{-6}$)	0.8306 (+126)	132/6 (0.8845)
DeepProZyme	`vanilla`	95/69,666	0.9022 (0.9331)	0.9010 (−85)	103/188 (1.0×10^{-6})	0.9017 (−33)	24/57 (3.2×10^{-4})	0.9027 (+36)	51/15 (0.9335)
DeepProZyme	$\leq 70\%$	95/69,666	0.8965 (0.9290)	0.8961 (−27)	151/178 (0.152)	0.8963 (−15)	44/59 (0.167)	0.8975 (+67)	78/11 (0.9296)
DeepProZyme	$\leq 50\%$	95/69,666	0.8966 (0.9288)	0.8966 (+1)	163/162 (1.000)	0.8966 (−3)	49/52 (0.842)	0.8979 (+87)	93/6 (0.9297)
DeepProZyme	$\leq 30\%$	95/69,666	0.8967 (0.9289)	0.8970 (+19)	170/151 (0.315)	0.8966 (−4)	51/55 (0.771)	0.8976 (+67)	76/9 (0.9296)
EnzBERT	`vanilla`	95/69,666	0.8430 (0.8707)	0.8416 (−92)	100/192 ($< 10^{-6}$)	0.8428 (−10)	51/61 (0.395)	0.8438 (+59)	65/6 (0.8713)
EnzBERT	$\leq 70\%$	95/69,666	0.8394 (0.8675)	0.8374 (−146)	96/242 ($< 10^{-6}$)	0.8391 (−26)	41/67 (0.0157)	0.8405 (+70)	76/6 (0.8682)
EnzBERT	$\leq 50\%$	95/69,666	0.8330 (0.8602)	0.8304 (−181)	132/313 ($< 10^{-6}$)	0.8324 (−38)	31/69 (1.8×10^{-4})	0.8343 (+89)	93/4 (0.8611)
EnzBERT	$\leq 30\%$	95/69,666	0.8256 (0.8515)	0.8232 (−167)	108/275 ($< 10^{-6}$)	0.8254 (−18)	39/57 (0.082)	0.8267 (+78)	80/2 (0.8524)

Table S5. 108-genome stoichiometric invariant audit (`results/pillar2_invariant_audit_summary.tsv`), curated BiGG reference GEM comparison ($n = 6$ species; `results/pillar2_curated6_gems_real.tsv`), and role-stratified `Price-149` CLEAN refinement across Strata A, B, and C (`results/pillar2_price149_by_stratum.tsv`).

Panel / Stratum	Method / Reconstruction Pipeline	Active / Growing	Added Rxns (Cert. / Pad.)	Imbal. / Dead-End Added	EC Prec. / Rec. / F_1	BiGG Rxn F_1	Top-1 Hits (Acc.)
108-Genome Audit	Raw Sequence Threshold (≥ 0.50)	2,809.5 (0% grow)	0.0 (0.0/0.0)	2,621.7 active / 9.75	n/a	n/a	n/a
	METEOR v8 (Negative-Cost MILP)	3,121.0 (100% grow)	313.35 (51.01/262.34)	2,597.8 active / 2.36	n/a	n/a	n/a
	Grow-Support Threshold Repair (Pos.-Cost LP)	2,584.8 (100% grow)	81.31 (81.31/0.00)	0.0 added / 2.23	n/a	n/a	n/a
	GSPA StoichiometricLocusRefiner (SLR)	2,815.2 (**300.6 flux**, 100% **grow**)	5.70 (5.70/0.00)	0.0 **added** / 0.00	n/a	n/a	n/a
6 Curated BiGG GEMs	CarveMe v1.5.2 (Native BiGG Database)	6/6 (100%)	n/a	n/a	0.8217/0.8927/0.8536	0.6955	n/a
	Reconstructor (ModelSEED Namespace)	4/6 (67%)	n/a	n/a	0.4387/0.8787/0.5844	n/a	n/a
	DeepProZyme + METEOR v8 (Neg.-Cost MILP)	6/6 (100%)	n/a	n/a	0.4383/0.8842/0.5851	0.2091	n/a
`Price-149` Stratum A ($n = 91$)	CLEAN Base → + SLR Cognate → + Single-Pred → + **Multi-Pred**	Core Biomass	n/a	n/a	$F_{\max}$: 0.6336 → 0.6405 → 0.6274 → **0.7019**	Macro: 0.4740 → **0.5288**	46 → 46 → 47 → **52** (57.1%)
`Price-149` Stratum B ($n = 55$)	CLEAN Base → + SLR Cognate → + Single-Pred → + **Multi-Pred**	Catabolic	n/a	n/a	$F_{\max}$: 0.3580 → 0.3753 → 0.3461 → **0.4546**	Macro: 0.3103 → **0.3436**	13 → 13 → **15** (27.3%) → 13 (23.6%)
`Price-149` Stratum C ($n = 2$)	CLEAN Base → + SLR Cognate → + Single-Pred → + **Multi-Pred**	Secondary	n/a	n/a	$F_{\max}$: 0.5099 → 0.5099 → 0.5175 → **0.5833**	Macro: 0.5000 → 0.5000	1 → 1 → 1 → 1 (50.0%)

Table S6. Summary of 5-fold genus-blocked cross-validation in `BacDive` across carbon-source growth ($n = 2{,}332$ assays, 529 strains) and diagnostic enzymes ($n = 17{,}496$ assays, 3,017 type strains; source: `results/pillar3_bacdive_per_fold_cv.tsv`).

Assay Panel	Classifier	Fold 0 F_1	Fold 1 F_1	Fold 2 F_1	Fold 3 F_1	Fold 4 F_1	5-Fold Mean $\pm$ s.d.	Pooled F_1 (BAcc / ROC-AUC)
Carbon Growth ($n = 2{,}332$)	BacDive-AI RF Baseline	0.8709	0.8720	0.8397	0.7174	0.8646	0.8329 ± 0.0659	0.8448 (0.8530/0.9222)
	DiASPora GBDT	0.8582	0.8819	0.8582	0.7817	0.8238	0.8408 ± 0.0390	0.8483 (0.8564/0.9268)
	Traitar L1-SVM	0.8480	0.8765	0.8727	0.7708	0.8653	0.8467 ± 0.0438	0.8571 (0.8638/0.9185)
	GSPA + DiASPora Hybrid	0.8830	0.9048	0.8824	0.7778	0.8602	0.8616 ± 0.0495	0.8710 (0.8783/0.9567)
Diagnostic Enzymes ($n = 17{,}496$)	BacDive-AI RF Baseline	0.8477	0.8371	0.8490	0.8553	0.8387	0.8456 ± 0.0076	0.8458 (0.7911/0.8593)
	DiASPora GBDT	0.8465	0.8579	0.8560	0.8655	0.8493	0.8550 ± 0.0075	0.8550 (0.8108/0.8699)
	Traitar L1-SVM	0.8620	0.8561	0.8562	0.8714	0.8517	0.8595 ± 0.0076	0.8593 (0.8014/0.8715)
	GSPA + DiASPora Hybrid	0.8671	0.8663	0.8718	0.8854	0.8653	0.8712 ± 0.0083	0.8713 (0.8257/0.9097)

S5 Stage 3 neuro-symbolic agent tool schemas, multi-seed stability, and extremophile case study

Stage 3 provides five deterministic, read-only genomic and ontological inspection tools to the open-weight instruction model `Qwen/Qwen3.8-27B-FP8` (`benchmark/agent/run_gspa_agent.py`):

1. `inspect_operon_neighborhood(protein_id, window=5)`: Returns the de-identified genomic neighborhood

(±5 genes on the same contig), intergenic distances, strand orientation, and neighbor GO/EC annotations.

2. `inspect_pathway_or_complex_gap(anomaly_id)`: Returns the target pathway or heteromeric complex identifier, fraction of partner steps/subunits present in the genome at $P \geq 0.50$, and list of missing steps/subunits.

3. `query_protein_evidence(protein_id)`: Returns Stage 0 per-channel calibrated probabilities, Pfam/InterPro structural domain annotations, and precomputed RLGC kernel density Bayes factors ($\mathrm{BF}_{\mathrm{operon}}$).

4. `check_symbolic_constraints(go_or_ec_term, taxon_id)`: Queries the deterministic ontology bridge (26,911 direct GO taxon rules and 2,715 GO–Uberon/PO/CL anatomical rules linked to 8,331 taxon constraints [4,767 `only_in_taxon`, 3,564 `never_in_taxon`] across 7,722 anatomical and 1,818 taxonomic hierarchy edges) and returns whether the term is valid in `taxon_id`, along with its nearest valid non-root GO ancestors if forbidden.

5. `propose_resolution(action, target_protein_id, resolved_term, confidence, justification)`: Submits the final structured decision (`PROMOTE`, `RELAX`, `DEMOTE`, or `ABSTAIN`), which is deterministically verified by `SymbolicGuardrail` before acceptance.

Supplementary Table S7 reports the per-seed and 3-seed summary of Config 3 (`Qwen/Qwen3.8-27B-FP8 + SymbolicGuardrail`) across seeds 42, 43, and 44 on the 120-anomaly and 50-unique-protein LRO benchmarks (source: `results/pillar4_multiseed_summary.tsv`), and Supplementary Table S8 summarizes the five desert extremophile isolates from the Empty Quarter (*Rub' al Khali*; source: `results/pillar6_empty_quarter_vignette.tsv`).

Table S7. Three-seed evaluation of Stage 3 Config 3 (`Qwen/Qwen3.8-27B-FP8 + SymbolicGuardrail`; source: `results/pillar4_multiseed_summary.tsv`).

Evaluation Run	Random Seed	Precision	Recall	F_1 Score	Accuracy	Violations Accepted	LRO-50 Hit@1 (MRR)
Stage 3 Config 3 (120 Anomalies + LRO-50)	Seed 42	0.9787	0.8519	0.9109	0.9250	0/120 (0.0%)	82.00% (0.8740)
Stage 3 Config 3 (120 Anomalies + LRO-50)	Seed 43	0.9592	0.8704	0.9126	0.9333	0/120 (0.0%)	84.00% (0.8890)
Stage 3 Config 3 (120 Anomalies + LRO-50)	Seed 44	0.9787	0.8519	0.9109	0.9250	0/120 (0.0%)	82.00% (0.8747)
Stage 3 Config 3 (3-Seed Mean ± s.d.)	**Seeds 42, 43, 44**	0.9722 ± 0.0092	0.8581 ± 0.0087	0.9115 ± 0.0008	0.9278 ± 0.0039	$0.00 \pm 0.00\%$	$82.67 \pm 0.94\%$ (0.8792 ± 0.0069)

Table S8. Empty Quarter (*Rub' al Khali*) desert extremophile isolate annotation summary across five bacterial genomes (source: `results/pillar6_empty_quarter_vignette.tsv`).

Empty Quarter Isolate	Taxonomic Clade	Proteins (N)	Raw C_{ess}	Stage 0 C_{ess}	GSPA C_{ess}	Path. Coh. (Raw → GSPA)	pFBA Repairs / Stage 3 Prom.
`EQ_01_Salinicola_sp`	Gammaproteobacteria	3,842	0.8810	0.9120	0.9680	0.4820 → 0.6140	6 repairs / 5 promotions
`EQ_02_Halomonas_sp`	Gammaproteobacteria	4,118	0.8920	0.9210	0.9740	0.4950 → 0.6280	5 repairs / 6 promotions
`EQ_03_Pseudomonas_sp`	Gammaproteobacteria	5,490	0.9140	0.9380	0.9820	0.5190 → 0.6410	4 repairs / 4 promotions
`EQ_04_Bacillus_endophyticus_EQ`	Bacillota	4,625	0.9010	0.9280	0.9760	0.5040 → 0.6320	5 repairs / 7 promotions
`EQ_05_Streptomyces_arid_EQ`	Actinomycetota	7,184	0.8540	0.8960	0.9590	0.4460 → 0.5980	8 repairs / 9 promotions
5-Isolate Mean	**3 Phyla / Classes**	5,051.8	0.8884	0.9190	0.9718	0.4892 → 0.6226	5.6 **repairs** / 6.2 **promotions**